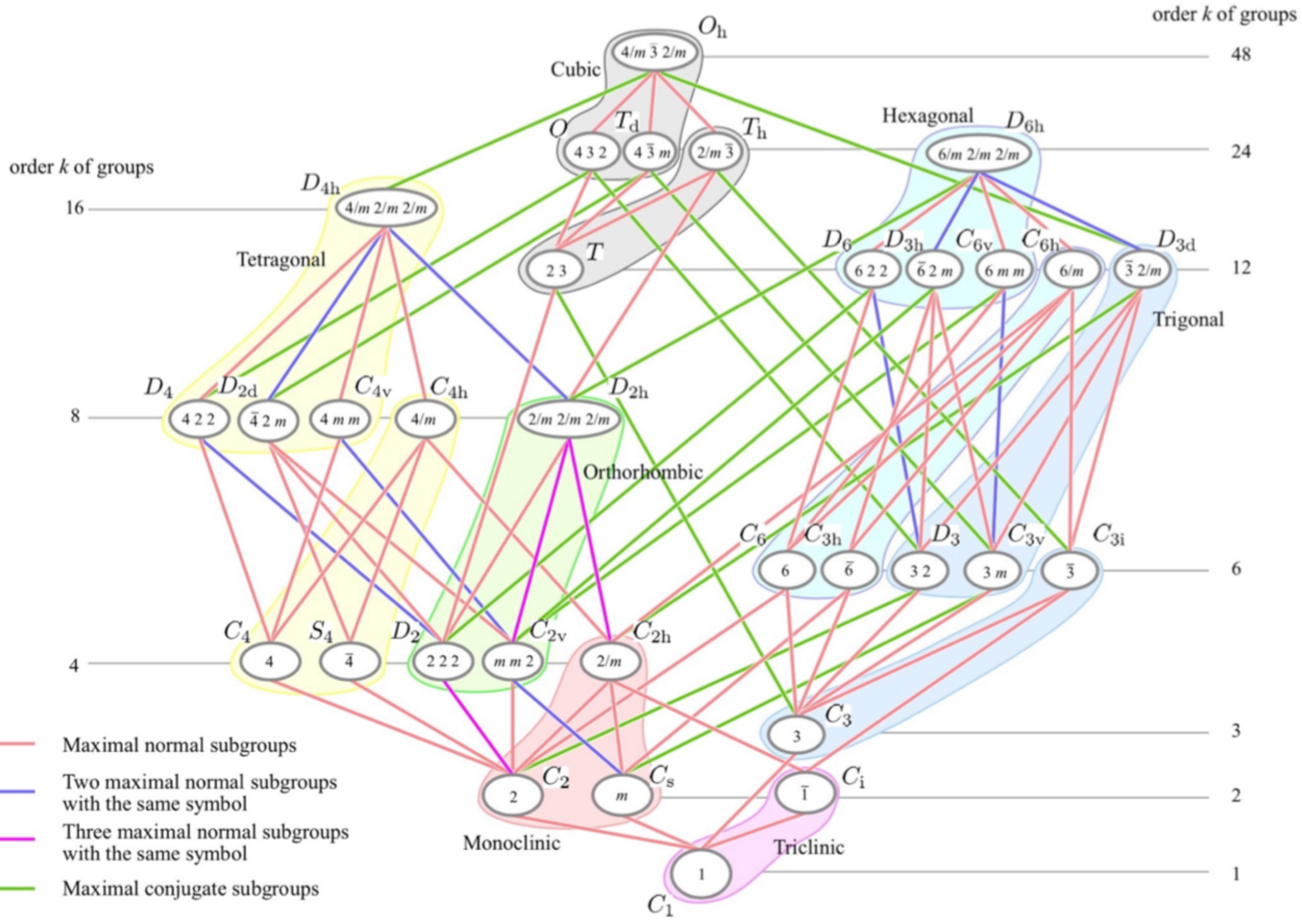
Compatibility Relation of Point Group

http://pmsl.planet.sci.kobe-u.ac.jp/~seto/class/V/supplement3-3.pdf

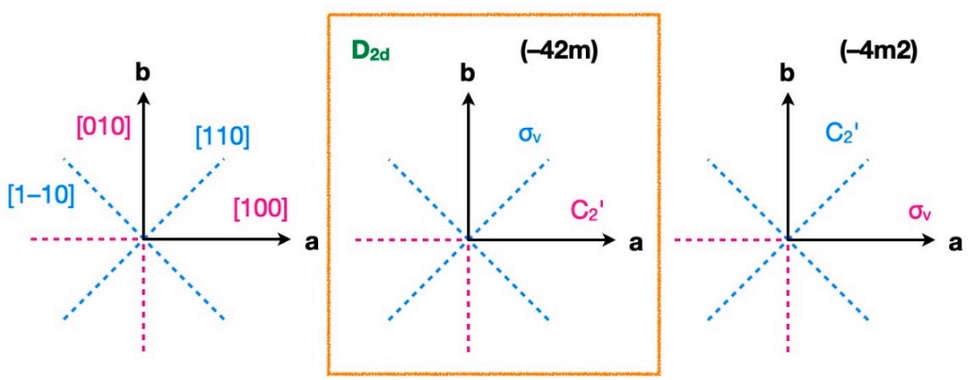


Multipole and Compatibility Relation

O_h Series

Table X. Multipoles under cubic, tetragonal, orthorhombic, and monoclinic crystals.[17] The upper and lower columns represent even-parity and odd-parity multipoles, respectively. The x axis is taken as the C_2' rotation, but the [110] axis is taken as the C_2' rotation for D_{2d}'.

E	ET	M	MT	O_h	O	T_d	T_h	T	D_{4h}	D_4	C_{4h}	D_{2d}	D_{2d}'	C_{4v}	C_4	S_4	D_{2h}	D_2	C_{2v}	C_{2h}	C_2	C_s
Q_0, Q_4	—	—	T_0, T_4	A_{1g}	A_1	A_1	A_g	A	A_{1g}	A_1	A_g	A_1	A_1	A_1	A	A	A_g	A	A_1	A_g	A	A′
—	G_{xyz}	M_{xyz}	—	A_{2g}	A_2	A_2	A_g	A	B_{1g}	B_1	B_g	B_1	B_2	B_1	B	B	A_g	A	A_1	A_g	A	A′
Q_u, Q_{4u}	—	—	T_u, T_{4u}	E_g	E	E	E_g	E	A_{1g}	A_1	A_g	A_1	A_1	A_1	A	A	A_g	A	A_1	A_g	A	A′
Q_v, Q_{4v}	—	—	T_v, T_{4v}						B_{1g}	B_1	B_g	B_1	B_2	B_1	B	B	A_g	A	A_1	A_g	A	A′
Q_{4x}^{α}	G_x, G_x^{α}	M_x, M_x^{α}	T_{4x}^{α}	T_{1g}	T_1	T_1	T_g	T	E_g	E	E_g	E	E	E	E	E	B_{3g}	B_3	B_2	B_g	B	A″
Q_{4y}^{α}	G_y, G_y^{α}	M_y, M_y^{α}	T_{4y}^{α}														B_{2g}	B_2	B_1	B_g	B	A″
Q_{4z}^{α}	G_z, G_z^{α}	M_z, M_z^{α}	T_{4z}^{α}						A_{2g}	A_2	A_g	A_2	A_2	A_2	A	A	B_{1g}	B_1	A_2	A_g	A	A′
Q_{yz}, Q_{4x}^{β}	G_x^{β}	M_x^{β}	T_{yz}, T_{4x}^{β}	T_{2g}	T_2	T_2	T_g	T	E_g	E	E_g	E	E	E	E	E	B_{3g}	B_3	B_2	B_g	B	A″
Q_{zx}, Q_{4y}^{β}	G_y^{β}	M_y^{β}	T_{zx}, T_{4y}^{β}														B_{2g}	B_2	B_1	B_g	B	A″
Q_{xy}, Q_{4z}^{β}	G_z^{β}	M_z^{β}	T_{xy}, T_{4z}^{β}						B_{2g}	B_2	B_g	B_2	B_1	B_2	B	B	B_{1g}	B_1	A_2	A_g	A	A′
—	G_0, G_4	M_0, M_4	—	A_{1u}	A_1	A_2	A_u	A	A_{1u}	A_1	A_u	B_1	B_1	A_2	A	B	A_u	A	A_2	A_u	A	A″
Q_{xyz}	—	—	T_{xyz}	A_{2u}	A_2	A_1	A_u	A	B_{1u}	B_1	B_u	A_1	A_2	B_2	B	A	A_u	A	A_2	A_u	A	A″
—	G_u, G_{4u}	M_u, M_{4u}	—	E_u	E	E	E_u	E	A_{1u}	A_1	A_u	B_1	B_1	A_2	A	B	A_u	A	A_2	A_u	A	A″
—	G_v, G_{4v}	M_v, M_{4v}	—						B_{1u}	B_1	B_u	A_1	A_2	B_2	B	A	A_u	A	A_2	A_u	A	A″
Q_x, Q_x^{α}	G_{4x}^{α}	M_{4x}^{α}	T_x, T_x^{α}	T_{1u}	T_1	T_2	T_u	T	E_u	E	E_u	E	E	E	E	E	B_{3u}	B_3	B_1	B_u	B	A′
Q_y, Q_y^{α}	G_{4y}^{α}	M_{4y}^{α}	T_y, T_y^{α}														B_{2u}	B_2	B_2	B_u	B	A′
Q_z, Q_z^{α}	G_{4z}^{α}	M_{4z}^{α}	T_z, T_z^{α}						A_{2u}	A_2	A_u	B_2	B_2	A_1	A	B	B_{1u}	B_1	A_1	A_u	A	A″
Q_x^{β}	G_{yz}, G_{4x}^{β}	M_{yz}, M_{4x}^{β}	T_x^{β}	T_{2u}	T_2	T_1	T_u	T	E_u	E	E_u	E	E	E	E	E	B_{3u}	B_3	B_1	B_u	B	A′
Q_y^{β}	G_{zx}, G_{4y}^{β}	M_{zx}, M_{4y}^{β}	T_y^{β}														B_{2u}	B_2	B_2	B_u	B	A′
Q_z^{β}	G_{xy}, G_{4z}^{β}	M_{xy}, M_{4z}^{β}	T_z^{β}						B_{2u}	B_2	B_u	A_2	A_1	B_1	B	A	B_{1u}	B_1	A_1	A_u	A	A″



-42m (D2d) -4m2 (D2d-1)

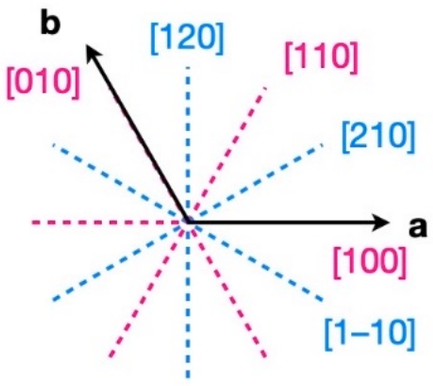
Standard setting
- monoclinic : b (y) axis (cell-choice 1)
- trigonal : hexagonal axes
- orthorhombic, tetragonal, cubic (origin choice 2 = inversion center)

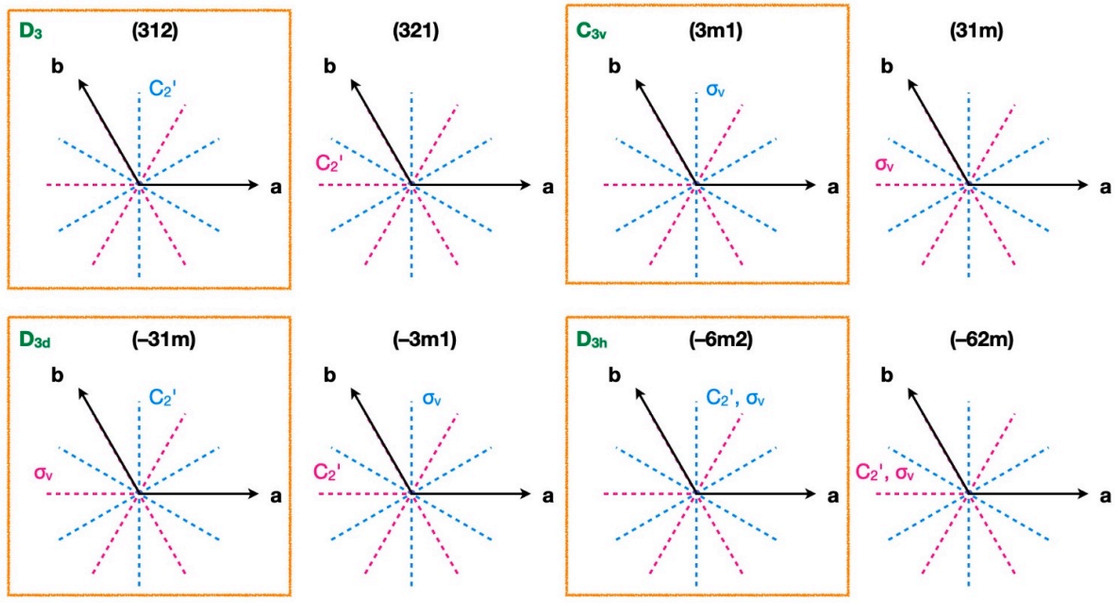
Multipole and Compatibility Relation

D_{6h} Series

Table XI. Multipoles under hexagonal and trigonal crystals.[17] The upper and lower columns represent even-parity and odd-parity multipoles, We take the x axis as the C_2' rotation axis; for D_{3h}', D_{3d}', and D_3', the C_2' rotation axis is taken along the y axis. For C_{3v} (C_{3v}'), we take the zx (yz) plane as the σ_v mirror plane. $C_{3i} = S_6$.

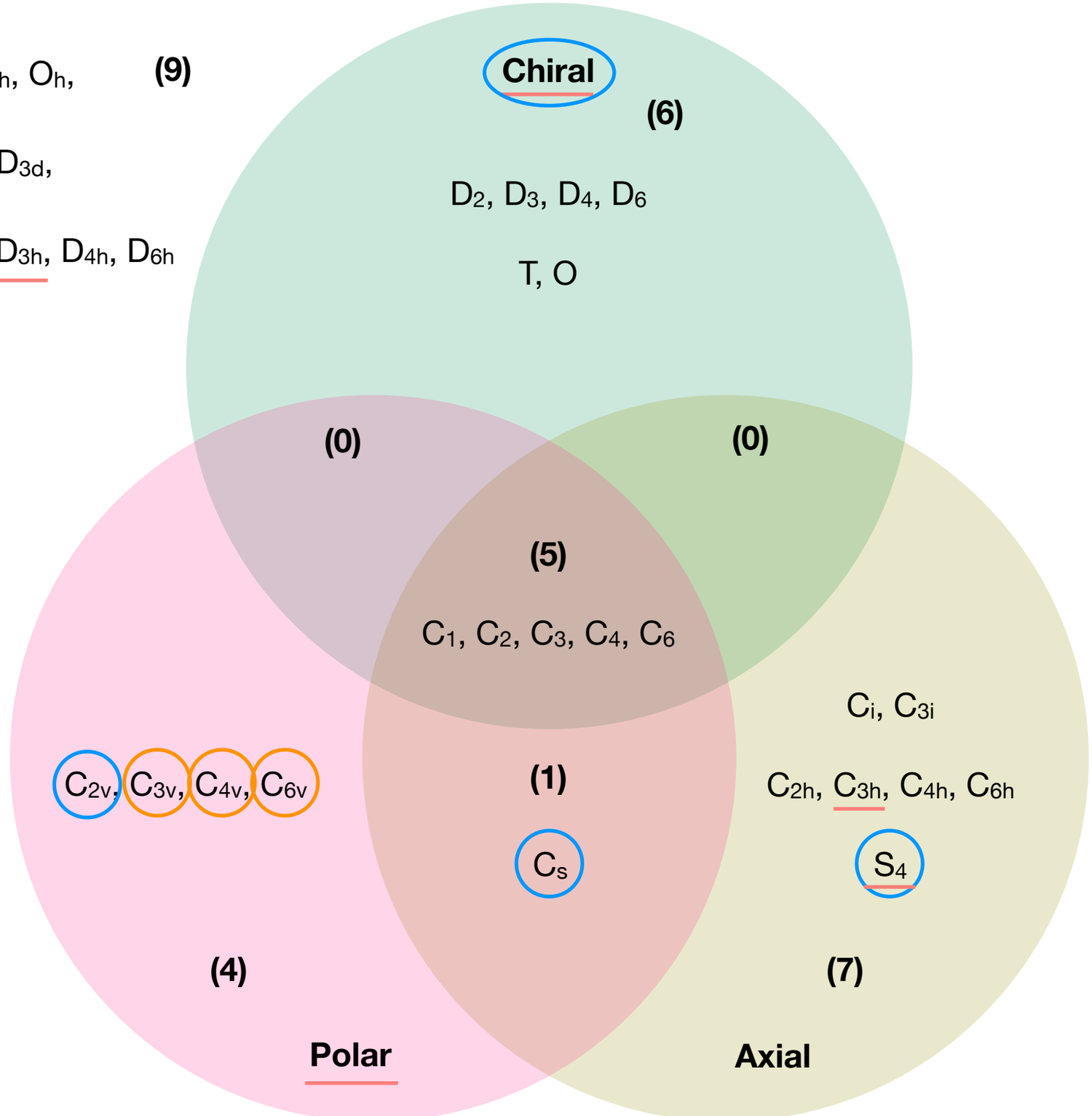
E	ET	M	MT	D_{6h}	D_6	C_{6h}	C_{6v}	C_6	D_{3h}	D_{3h}'	C_{3h}	D_{3d}	D_{3d}'	D_3	D_3'	C_{3v}	C_{3v}'	C_{3i}	C_3
Q_0, Q_u, Q_{40}	—	—	T_0, T_u, T_{40}	A_{1g}	A_1	A_g	A_1	A	A_1'	A_1'	A'	A_{1g}	A_{1g}	A_1	A_1	A_1	A_1	A_g	A
—	G_z, G_z^α	M_z, M_z^α	—	A_{2g}	A_2	A_g	A_2	A	A_2'	A_2'	A'	A_{2g}	A_{2g}	A_2	A_2	A_2	A_2	A_g	A
Q_{4a}	G_{3a}	M_{3a}	T_{4a}	B_{1g}	B_1	B_g	B_2	B	A_1''	A_2''	A''	A_{1g}	A_{2g}	A_1	A_2	A_2	A_1	A_g	A
Q_{4b}	G_{3b}	M_{3b}	T_{4b}	B_{2g}	B_2	B_g	B_1	B	A_2''	A_1''	A''	A_{2g}	A_{1g}	A_2	A_1	A_1	A_2	A_g	A
Q_{zx}, Q_{4u}^α	G_x, G_{3u}	M_x, M_{3u}	T_{zx}, T_{4u}^α	E_{1g}	E_1	E_{1g}	E_1	E_1	E''	E''	E''	E_g	E_g	E	E	E	E	E_g	E
Q_{yz}, Q_{4v}^α	G_y, G_{3v}	M_y, M_{3v}	T_{yz}, T_{4v}^α																
Q_v, $Q_{4u}^{\beta1}$, $Q_{4u}^{\beta2}$	G_{xyz}	M_{xyz}	T_v, $T_{4u}^{\beta1}$, $T_{4u}^{\beta2}$	E_{2g}	E_2	E_{2g}	E_2	E_2	E'	E'	E'	E_g	E_g	E	E	E	E	E_g	E
Q_{xy}, $Q_{4v}^{\beta1}$, $Q_{4v}^{\beta2}$	G_z^β	M_z^β	T_{xy}, $T_{4v}^{\beta2}$, $T_{4v}^{\beta2}$																
—	G_0, G_u, G_{40}	M_0, M_u, M_{40}	—	A_{1u}	A_1	A_u	A_2	A	A_1''	A_1''	A''	A_{1u}	A_{1u}	A_1	A_1	A_2	A_2	A_u	A
Q_z, Q_z^α	—	—	T_z, T_z^α	A_{2u}	A_2	A_u	A_1	A	A_2''	A_2''	A''	A_{2u}	A_{2u}	A_2	A_2	A_1	A_1	A_u	A
Q_{3a}	G_{4a}	M_{4a}	T_{3a}	B_{1u}	B_1	B_u	B_1	B	A_1'	A_2'	A'	A_{1u}	A_{2u}	A_1	A_2	A_1	A_2	A_u	A
Q_{3b}	G_{4b}	M_{4b}	T_{3b}	B_{2u}	B_2	B_u	B_2	B	A_2'	A_1'	A'	A_{2u}	A_{1u}	A_2	A_1	A_2	A_1	A_u	A
Q_x, Q_{3u}	G_{zx}, G_{4u}^α	M_{zx}, M_{4u}^α	T_x, T_{3u}	E_{1u}	E_1	E_{1u}	E_1	E_1	E'	E'	E'	E_u	E_u	E	E	E	E	E_u	E
Q_y, Q_{3v}	G_{yz}, G_{4v}^α	M_{yz}, M_{4v}^α	T_y, T_{3v}																
Q_{xyz}	G_v, $G_{4u}^{\beta1}$, $G_{4u}^{\beta2}$	M_v, $M_{4u}^{\beta1}$, $M_{4u}^{\beta2}$	T_{xyz}	E_{2u}	E_2	E_{2u}	E_2	E_2	E''	E''	E''	E_u	E_u	E	E	E	E	E_u	E
Q_z^β	G_{xy}, $G_{4v}^{\beta1}$, $G_{4v}^{\beta2}$	M_{xy}, $M_{4v}^{\beta1}$, $M_{4v}^{\beta2}$	T_z^β																
									-62m (D_{3h}-1)	-6m2 (D_{3h})		-3m1 (D_{3d}-1)	-31m (D_{3d})	321 (D_3-1)	312 (D_3)	31m (C_{3v}-1)	3m1 (C_{3v})		





Classification of Point Group

Presence of time-reversal symmetry (= Gray point group)



Polar (E-dipole)

Q_x, Q_y, Q_z

Axial (ET-dipole)

G_x, G_y, G_z

Chiral (ET-monopole)

G_0

weak Gyrotropic

no G_0 and G_2

strong Gyrotropic

G_0 or G_2

Noncentrosymmetric

odd-parity multipole

Classification of Noncentrosymmetric Point Group

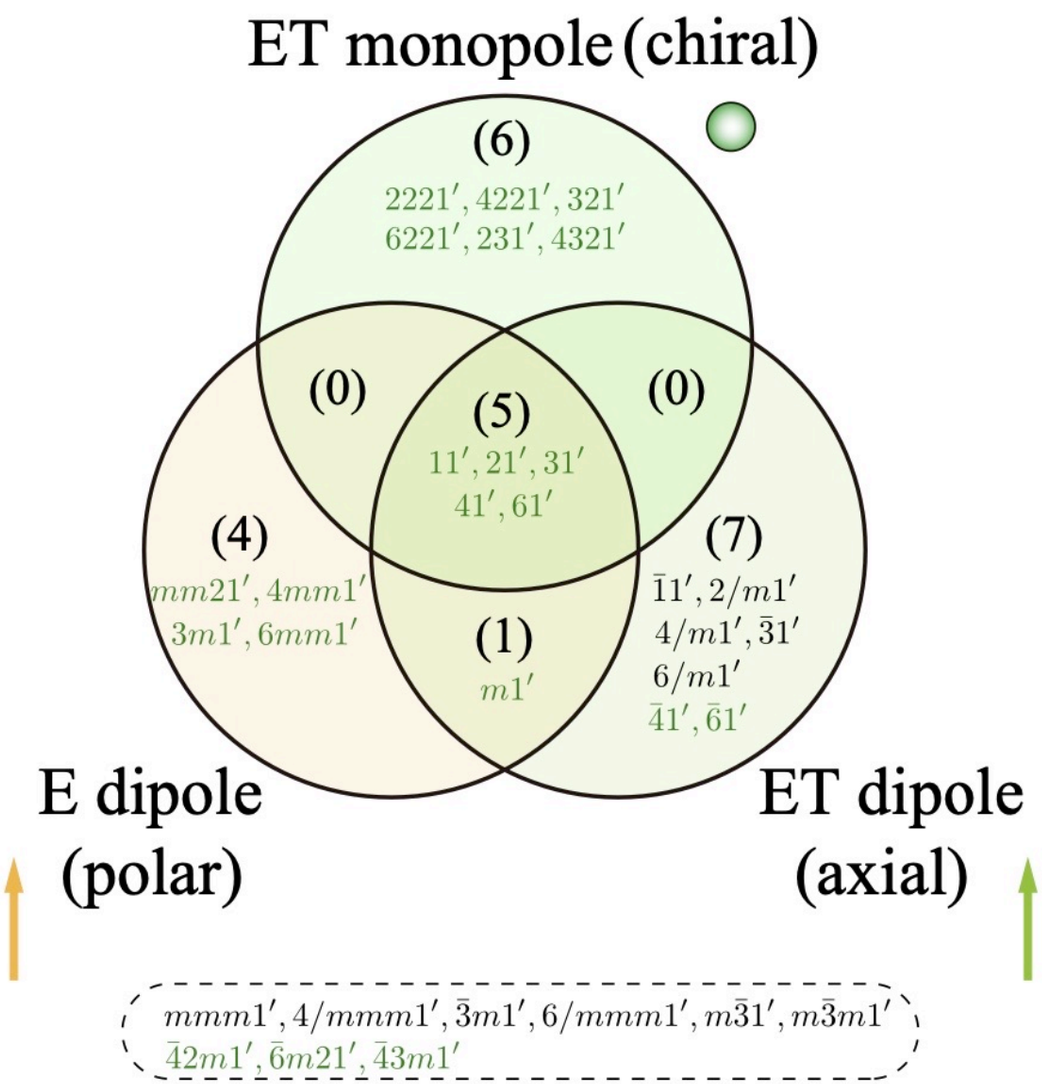
chiral | weak gyrotropic NOA inactive | pure strong gyrotropic NOA active | polar

crystal system	point group	N	P	C	G	totally symmetric rep. rank 0	rank 1	rank 2	rank 3	rank 4
cubic	O	✓		✓	✓	G_0				G_4
	T_d	✓							Q_{xyz}	
	T_h	✓		✓	✓	G_0			Q_{xyz}	G_4
tetragonal	D_4	✓		✓	✓	G_0		G_u		$G_4,\ G_{4u}$
	D_{2d}	✓			✓			G_v	Q_{xyz}	G_{4v}
	C_{4v}	✓	✓		✓		Q_z		Q_z^α	G_{4z}^α
C_{4v}, C_{4h}	C_4	✓	✓	✓	✓	G_0	Q_z	G_u	Q_z^α	$G_4,\ G_{4u},\ G_{4z}^\alpha$
	S_4	✓			✓			$G_v,\ G_{xy}$	$Q_{xyz},\ Q_z^\beta$	$G_{4v},\ G_{4z}^\beta$
orthorhombic	D_2	✓		✓	✓	G_0		$G_u,\ G_v$	Q_{xyz}	$G_4,\ G_{4u},\ G_{4v}$
D_{2d}, D_{2h}	C_{2v}	✓	✓		✓		Q_z	G_{xy}	$Q_z^\alpha,\ Q_z^\beta$	$G_{4z}^\alpha,\ G_{4z}^\beta$
monoclinic C_{2h}	C_2	✓	✓	✓	✓	G_0	Q_z	$G_u,\ G_v,\ G_{xy}$	$Q_{xyz},\ Q_z^\alpha,\ Q_z^\beta$	$G_4,\ G_{4u},\ G_{4v},\ G_{4z}^\alpha,\ G_{4z}^\beta$
principal z axis	C_s	✓	✓		✓		$Q_x,\ Q_y$	$G_{yz},\ G_{zx}$	$Q_x^\alpha,\ Q_y^\alpha,\ Q_x^\beta,\ Q_y^\beta$	$G_{4x}^\alpha,\ G_{4y}^\alpha,\ G_{4x}^\beta,\ G_{4y}^\beta$
hexagonal D_{6h}	D_6	✓		✓	✓	G_0		G_u		G_{40}
	D_{3h}	✓							Q_{3a}	G_{4a}
	C_{6v}	✓	✓		✓		Q_z		Q_z^α	
C_{6v}, C_{6h}	C_6	✓	✓	✓	✓	G_0	Q_z	G_u	Q_z^α	G_{40}
	C_{3h}	✓							$Q_{3a},\ Q_{3b}$	$G_{4a},\ G_{4b}$
trigonal D_{3h}, D_{3d}	D_3	✓		✓	✓	G_0		G_u	Q_{3a}	$G_{40},\ G_{4a}$
	C_{3v}	✓	✓		✓		Q_z		$Q_z^\alpha,\ Q_{3a}$	G_{4a}
C_{3h}, C_{3v}, C_{3i}	C_3	✓	✓	✓	✓	G_0	Q_z	G_u	$Q_z^\alpha,\ Q_{3a},\ Q_{3b}$	$G_{40},\ G_{4a},\ G_{4b}$

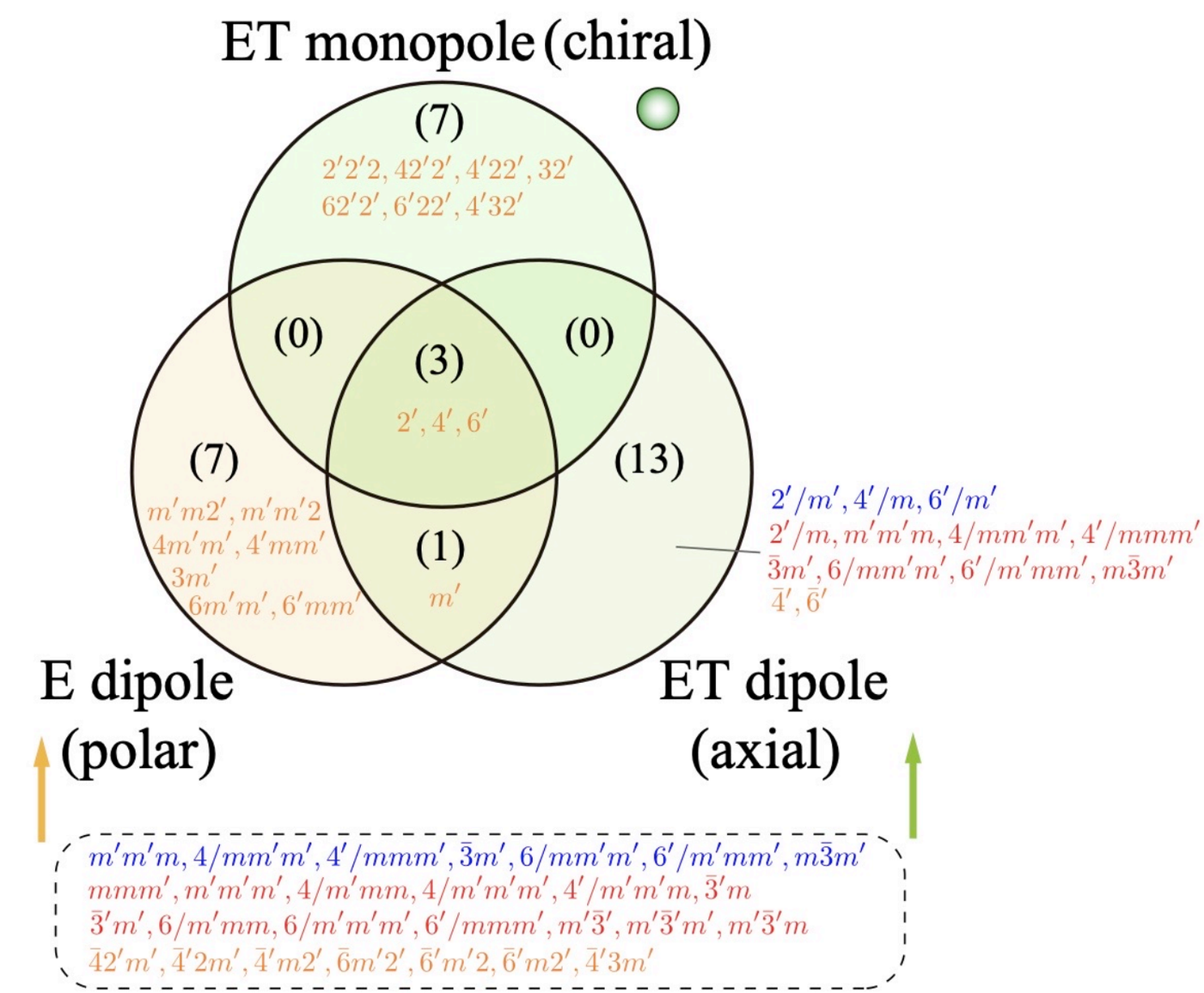
Active Multipole in Magnetic Point Group

Electric Multipole

Gray point group (GPG)

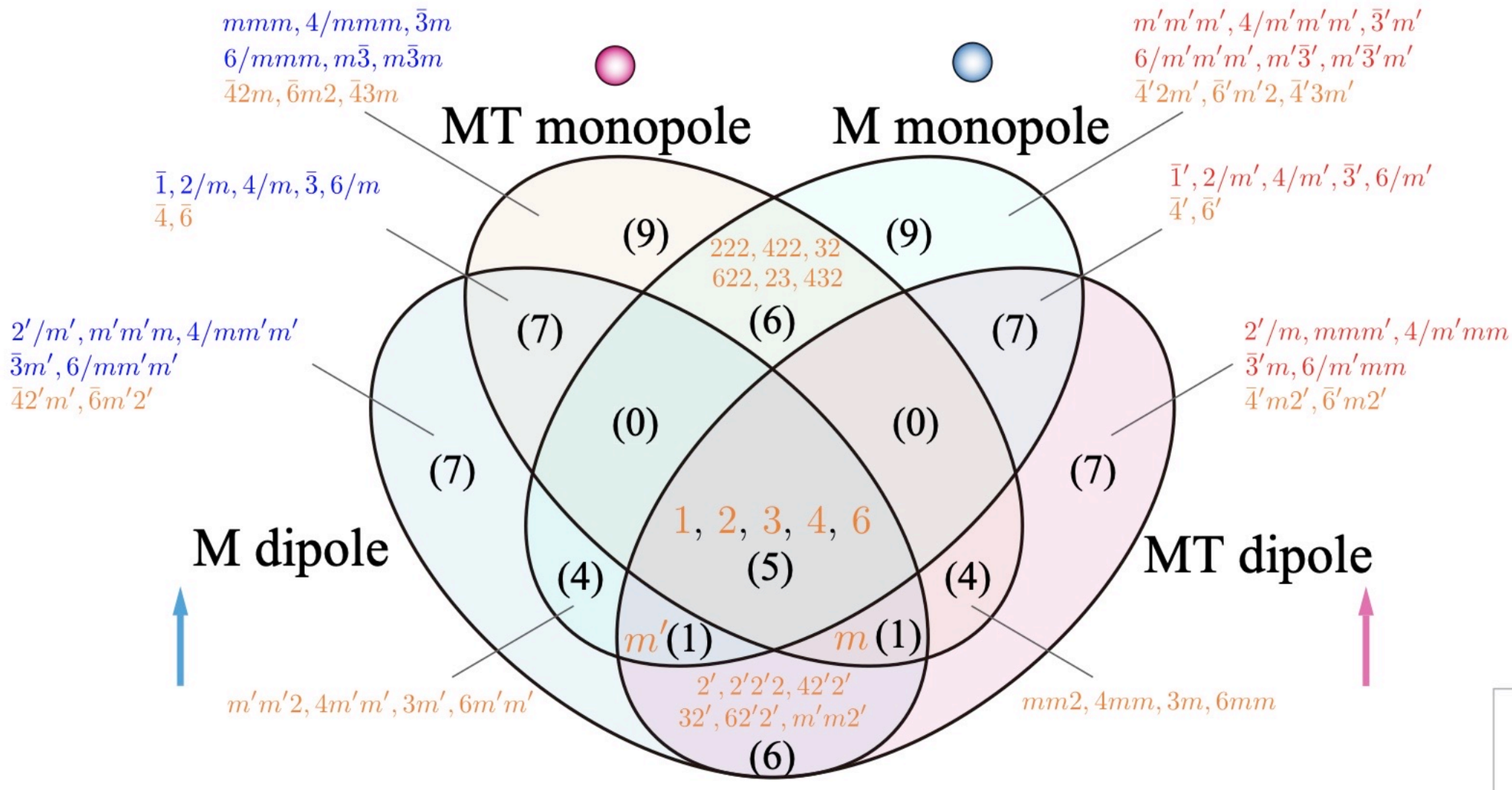


Black-and-white point group (BWPG)



Active Multipole in Magnetic Point Group

Magnetic Multipole



GPG

$\bar{1}1', 2/m1', mmm1', 4/m1', 4/mmm1', \bar{3}1', \bar{3}m1'$
$6/m1', 6/mmm1', m\bar{3}1', m\bar{3}m1'$
$11', 21', m1', 2221', mm21', 41', \bar{4}1', 4221', 4mm1', \bar{4}2m1'$
$31', 321', 3m1', 61', \bar{6}1', 6221', 6mm1', \bar{6}m21', 231', \bar{4}3m1', 4321'$

BWPG

$4'/m, 4'/mmm', 6'/m', 6'/m'mm', m\bar{3}m'$
$4'/m', 4'/m'm'm, 6'/m, 6'/mmm', m'\bar{3}'m$
$4', 4'22', 4'mm', 6', 6'22', 6'mm', 4'32'$

$(\mathcal{P}, \mathcal{T}) = (\bigcirc, \bigcirc)$
$(\mathcal{P}, \mathcal{T}) = (\times, \bigcirc)$
$(\mathcal{P}, \mathcal{T}) = (\bigcirc, \times)$
$(\mathcal{P}, \mathcal{T}, \mathcal{PT}) = (\times, \times, \bigcirc)$
$(\mathcal{P}, \mathcal{T}, \mathcal{PT}) = (\times, \times, \times)$

Antisymmetric Spin-Orbit Coupling

Table V. Functional forms of the ASOC for the periodic crystal and $\boldsymbol{k} \to \boldsymbol{0}$ limit and under noncentrosymmetric point groups (PGs). For $D_{2\mathrm{d}}$, $D_{3\mathrm{h}}$, $C_{3\mathrm{v}}$, and D_3, the setting of $-42m$, $-6m2$, $3m1$, and 312 is adopted, respectively. The principle axis for the monoclinic crystal is taken along the y axis. We use the abbreviation like $\sin k_x \to s_x$ and $\cos k_y/2 \to c_{y/2}$ for notational simplicity. $f_x = \frac{2}{\sqrt{3}}\left(2c_{x/2}+c_{\sqrt{3}y/2}\right)s_{x/2}$, $f_y = 2c_{x/2}s_{\sqrt{3}y/2}$, $f'_x = \frac{4}{\sqrt{6}}\left(-c_x + c_{x/2}\cos_{\sqrt{3}y/2}\right)s_z$, $f'_y = -2\sqrt{2}s_{x/2}s_{\sqrt{3}y/2}s_z$, $f_1 = \frac{4}{\sqrt{6}}\left(-c_{3x/2}+c_{\sqrt{3}y/2}\right)s_{\sqrt{3}y/2}$, $f_2 = \frac{4}{\sqrt{6}}\left(-c_{x/2}+c_{\sqrt{3}y/2}\right)s_{x/2}$.

PG	IRREP.	ASOC for periodic crystal and $\boldsymbol{k} \to \boldsymbol{0}$ limit	active PG	multipole	notes
O_{h}	A_{1u}	$s_x\sigma_x + s_y\sigma_y + s_z\sigma_z \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} k_x\sigma_x + k_y\sigma_y + k_z\sigma_z$	O, T	G_0	hedgehog
	A_{2u}	$(-c_y+c_z)s_x\sigma_x + (-c_z+c_x)s_y\sigma_y + (-c_x+c_y)s_z\sigma_z$ $\underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} (k_y^2-k_z^2)\sigma_x + (k_z^2-k_x^2)\sigma_y + (k_z^2-k_x^2)\sigma_y$	T_{d}, T	Q_{xyz}	Dresselhaus
$D_{4\mathrm{h}}$	A_{1u}	$c_1(s_x\sigma_x + s_y\sigma_y) + c_2 s_z\sigma_z \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1(k_x\sigma_x + k_y\sigma_y) + c_2 k_z\sigma_z$	D_4, C_4	G_0, G_u	hedgehog
	A_{2u}	$s_y\sigma_x - s_x\sigma_y \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} k_y\sigma_x - k_x\sigma_y$	$C_{4\mathrm{v}}$, C_4	Q_z	Rashba
	B_{1u}	$s_x\sigma_x - s_y\sigma_y \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} k_x\sigma_x - k_y\sigma_y$	$D_{2\mathrm{d}}$, S_4	G_v	v type
	B_{2u}	$s_y\sigma_x + s_x\sigma_y \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} k_y\sigma_x + k_x\sigma_y$	S_4	G_{xy}	xy type
$D_{2\mathrm{h}}$	A_u	$c_1 s_x\sigma_x + c_2 s_y\sigma_y + c_3 s_z\sigma_z \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1 k_x\sigma_x + c_2 k_y\sigma_y + c_3 k_z\sigma_z$	D_2	G_0, G_u, G_v	
	B_{1u}	$c_1 s_y\sigma_x + c_2 s_x\sigma_y + c_3 s_x s_y s_z\sigma_z$ $\underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1 k_y\sigma_x + c_2 k_x\sigma_y + c_3 k_x k_y k_z\sigma_z$	$C_{2\mathrm{v}}$	Q_z, G_{xy}, Q_z^{β}	
$C_{2\mathrm{h}}$	A_u	$(c_1 s_x + c_2 s_z)\sigma_x + c_3 s_y\sigma_y + (c_4 s_x + c_5 s_z)\sigma_z$ $\underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} (c_1 k_x + c_2 k_z)\sigma_x + c_3 k_y\sigma_y + (c_4 k_x + c_5 k_z)\sigma_z$	C_2	$G_0, Q_z, G_u, G_v, G_{zx}$	
	B_u	$c_1 s_y\sigma_x + (c_2 s_x + c_3 s_z)\sigma_y + c_4 s_y\sigma_z$ $\underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1 k_y\sigma_x + (c_2 k_x + c_3 k_z)\sigma_y + c_4 k_y\sigma_z$	C_{s}	Q_x, Q_z, G_{yz}, G_{xy}	
C_{i}	A_u	$(c_1 s_x + c_2 s_y + c_3 s_z)\sigma_x + (c_4 s_x + c_5 s_y + c_6 s_z)\sigma_y$ $+(c_7 s_x + c_8 s_y + c_9 s_z)\sigma_z$ $\underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} (c_1 k_x + c_2 k_y + c_3 k_z)\sigma_x + (c_4 k_x + c_5 k_y + c_6 k_z)\sigma_y$ $+(c_7 k_x + c_8 k_y + c_9 k_z)\sigma_z$	C_1	G_0, Q_x, Q_y, Q_z $G_u, G_v, G_{yz}, G_{zx}, G_{xy}$	
$D_{6\mathrm{h}}$	A_{1u}	$c_1(f_x\sigma_x + f_y\sigma_y) + c_2 s_z\sigma_z \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1(k_x\sigma_x + k_y\sigma_y) + c_2 k_z\sigma_z$	D_6, C_6	G_0, G_u	hedgehog
	A_{2u}	$f_y\sigma_x - f_x\sigma_y \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} k_y\sigma_x - k_x\sigma_y$	$C_{6\mathrm{v}}$, C_6	Q_z	Rashba
	B_{1u}	$c_1(f'_x\sigma_x + f'_y\sigma_y) + c_2 f_2\sigma_z$ $\underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1\left[(k_x^2-k_y^2)k_z\sigma_x - k_x k_y k_z\sigma_y\right] + c_2(k_x^2-3k_y^2)k_x\sigma_z$	$D_{3\mathrm{h}}$, $C_{3\mathrm{h}}$	Q_{3b}, G_{4b}	
	B_{2u}	$c_1(f'_y\sigma_x - f'_x\sigma_y) + c_2 f_1\sigma_z \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1(f'_y\sigma_x - f'_x\sigma_y) + c_2 f_1\sigma_z$ $\underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1\left[-k_x k_y k_z\sigma_x - (k_x^2-k_y^2)k_z\sigma_y\right] + c_2(3k_x^2-k_y^2)k_y\sigma_z$	$C_{3\mathrm{h}}$	Q_{3a}, G_{4a}	
$D_{3\mathrm{d}}$	A_{1u}	$c_1(f_x\sigma_x + f_y\sigma_y) + c_2 s_z\sigma_z \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} c_1(k_x\sigma_x + k_y\sigma_y) + c_2 k_z\sigma_z$	D_3, C_3	G_0, G_u	hedgehog
	A_{2u}	$f_y\sigma_x - f_x\sigma_y \underset{(\boldsymbol{k}\to\boldsymbol{0})}{\to} k_y\sigma_x - k_x\sigma_y$	$C_{3\mathrm{v}}$, C_3	Q_z	Rashba

Compatibility Relation (from MultiPie)

chiral

weak gyrotropic
NOA inactive

pure strong gyrotropic
NOA active

polar

O_h	A_{1g}	A_{2g}	$E_g(x)$	$E_g(y)$	$T_{1g}(x)$	$T_{1g}(y)$	$T_{1g}(z)$	$T_{2g}(x)$	$T_{2g}(y)$	$T_{2g}(z)$	A_{1u}	A_{2u}	$E_u(x)$	$E_u(y)$	$T_{1u}(x)$	$T_{1u}(y)$	$T_{1u}(z)$	$T_{2u}(x)$	$T_{2u}(y)$	$T_{2u}(z)$
Q	s	i	du	dv	gax	gay	gaz	dyz	dxz	dxy	l	f3	hu	hv	px	py	pz	fbx	fby	fbz
G	l	f3	hu	hv	px	py	pz	fbx	fby	fbz	s	i	du	dv	gax	gay	gaz	dyz	dxz	dxy
O_h	A_{1g}	A_{2g}	$E_g(x)$	$E_g(y)$	$T_{1g}(x)$	$T_{1g}(y)$	$T_{1g}(z)$	$T_{2g}(x)$	$T_{2g}(y)$	$T_{2g}(z)$	A_{1u}	A_{2u}	$E_u(x)$	$E_u(y)$	$T_{1u}(x)$	$T_{1u}(y)$	$T_{1u}(z)$	$T_{2u}(x)$	$T_{2u}(y)$	$T_{2u}(z)$
T_d	A_1	A_2	$E(x)$	$E(y)$	$T_1(x)$	$T_1(y)$	$T_1(z)$	$T_2(x)$	$T_2(y)$	$T_2(z)$	A_2	A_1	$E(y)$	$-E(x)$	$T_2(x)$	$T_2(y)$	$T_2(z)$	$T_1(x)$	$T_1(y)$	$T_1(z)$
O	A_1	A_2	$E(x)$	$E(y)$	$T_1(x)$	$T_1(y)$	$T_1(z)$	$T_2(x)$	$T_2(y)$	$T_2(z)$	A_1	A_2	$E(x)$	$E(y)$	$T_1(x)$	$T_1(y)$	$T_1(z)$	$T_2(x)$	$T_2(y)$	$T_2(z)$
T_h	A_g	A_g	$E_g(x)$	$E_g(y)$	$T_g(x)$	$T_g(y)$	$T_g(z)$	$T_g(x)$	$T_g(y)$	$T_g(z)$	A_u	A_u	$E_u(x)$	$E_u(y)$	$T_u(x)$	$T_u(y)$	$T_u(z)$	$T_u(x)$	$T_u(y)$	$T_u(z)$
T_h(c)	A_g	A_g	$E_g^{(a)}$	$E_g^{(b)}$	$T_g(x)$	$T_g(y)$	$T_g(z)$	$T_g(x)$	$T_g(y)$	$T_g(z)$	A_u	A_u	$E_u^{(a)}$	$E_u^{(b)}$	$T_u(x)$	$T_u(y)$	$T_u(z)$	$T_u(x)$	$T_u(y)$	$T_u(z)$
T	A	A	$E(x)$	$E(y)$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$	A	A	$E(x)$	$E(y)$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$
T(c)	A	A	$E^{(a)}$	$E^{(b)}$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$	A	A	$E^{(a)}$	$E^{(b)}$	$T(x)$	$T(y)$	$T(z)$	$T(x)$	$T(y)$	$T(z)$
D_4h	A_{1g}	B_{1g}	A_{1g}	B_{1g}	$E_g(x)$	$-E_g(y)$	A_{2g}	$E_g(x)$	$E_g(y)$	B_{2g}	A_{1u}	B_{1u}	A_{1u}	B_{1u}	$E_u(x)$	$E_u(y)$	A_{2u}	$E_u(x)$	$-E_u(y)$	B_{2u}
D_2d	A_1	B_1	A_1	B_1	$E(x)$	$-E(y)$	A_2	$E(x)$	$E(y)$	B_2	B_1	A_1	B_1	A_1	$E(x)$	$E(y)$	B_2	$E(x)$	$-E(y)$	A_2
$D_\mathrm{2d}(1)$	A_1	B_2	A_1	B_2	$E(y)$	$E(x)$	A_2	$E(y)$	$-E(x)$	B_1	B_1	A_2	B_1	A_2	$E(x)$	$E(y)$	B_2	$E(x)$	$-E(y)$	A_1
C_4v	A_1	B_1	A_1	B_1	$E(y)$	$-E(x)$	A_2	$E(y)$	$E(x)$	B_2	A_2	B_2	A_2	B_2	$E(x)$	$E(y)$	A_1	$E(x)$	$-E(y)$	B_1
D_4	A_1	B_1	A_1	B_1	$E(x)$	$E(y)$	A_2	$E(x)$	$-E(y)$	B_2	A_1	B_1	A_1	B_1	$E(x)$	$E(y)$	A_2	$E(x)$	$-E(y)$	B_2
C_4h	A_g	B_g	A_g	B_g	$E_g(y)$	$E_g(x)$	A_g	$E_g(x)$	$E_g(y)$	B_g	A_u	B_u	A_u	B_u	$E_u(x)$	$E_u(y)$	A_u	$E_u(y)$	$E_u(x)$	B_u
C_4h(c)	A_g	B_g	A_g	B_g	$E_g^{(a)}$	$E_g^{(b)}$	A_g	$E_g^{(b)}$	$E_g^{(a)}$	B_g	A_u	B_u	A_u	B_u	$E_u^{(a)}$	$E_u^{(b)}$	A_u	$E_u^{(b)}$	$E_u^{(a)}$	B_u
S_4	A	B	A	B	$E(y)$	$E(x)$	A	$E(x)$	$E(y)$	B	B	A	B	A	$E(x)$	$E(y)$	B	$E(y)$	$E(x)$	A
S_4(c)	A	B	A	B	$E^{(b)}$	$E^{(a)}$	A	$E^{(a)}$	$E^{(b)}$	B	B	A	B	A	$E^{(a)}$	$E^{(b)}$	B	$E^{(b)}$	$E^{(a)}$	A
C_4	A	B	A	B	$E(x)$	$E(y)$	A	$E(y)$	$E(x)$	B	A	B	A	B	$E(x)$	$E(y)$	A	$E(y)$	$E(x)$	B
C_4(c)	A	B	A	B	$E^{(a)}$	$E^{(b)}$	A	$E^{(b)}$	$E^{(a)}$	B	A	B	A	B	$E^{(a)}$	$E^{(b)}$	A	$E^{(b)}$	$E^{(a)}$	B
D_2h	A_g	A_g	A_g	A_g	B_{3g}	B_{2g}	B_{1g}	B_{3g}	B_{2g}	B_{1g}	A_u	A_u	A_u	A_u	B_{3u}	B_{2u}	B_{1u}	B_{3u}	B_{2u}	B_{1u}
C_2v	A_1	A_1	A_1	A_1	B_2	B_1	A_2	B_2	B_1	A_2	A_2	A_2	A_2	A_2	B_1	B_2	A_1	B_1	B_2	A_1
D_2	A	A	A	A	B_3	B_2	B_1	B_3	B_2	B_1	A	A	A	A	B_3	B_2	B_1	B_3	B_2	B_1
C_2h	A_g	A_g	A_g	A_g	B_g	A_g	B_g	B_g	A_g	B_g	A_u	A_u	A_u	A_u	B_u	A_u	B_u	B_u	A_u	B_u
C_s	A'	A'	A'	A'	A''	A'	A''	A''	A'	A''	A''	A''	A''	A''	A'	A''	A'	A'	A''	A'
C_2	A	A	A	A	B	A	B	B	A	B	A	A	A	A	B	A	B	B	A	B
C_i	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_g	A_u	A_u	A_u	A_u	A_u	A_u	A_u	A_u	A_u	A_u
C_1	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A

Compatibility Relation (from MultiPie)

chiral

weak gyrotropic
NOA inactive

pure strong gyrotropic
NOA active

polar

D_{6h}	A_{1g}	A_{2g}	B_{1g}	B_{2g}	$E_{1g}(x)$	$E_{1g}(y)$	$E_{2g}(x)$	$E_{2g}(y)$	A_{1u}	A_{2u}	B_{1u}	B_{2u}	$E_{1u}(x)$	$E_{1u}(y)$	$E_{2u}(x)$	$E_{2u}(y)$
Q	s/du	i0	ga	gb	dyz	dxz'	dv	dxy'	j	pz	f1	f2	px	py	f3	fbz
G	j	pz	f1	f2	px	py	f3	fbz	s/du	i0	ga	gb	dyz	dxz'	dv	dxy'
D_{6h}	A_{1g}	A_{2g}	B_{1g}	B_{2g}	$E_{1g}(x)$	$E_{1g}(y)$	$E_{2g}(x)$	$E_{2g}(y)$	A_{1u}	A_{2u}	B_{1u}	B_{2u}	$E_{1u}(x)$	$E_{1u}(y)$	$E_{2u}(x)$	$E_{2u}(y)$
D_{3h}	A_1'	A_2'	A_1''	A_2''	$E''(x)$	$E''(y)$	$E'(y)$	$-E'(x)$	A_1''	A_2''	A_1'	A_2'	$E'(x)$	$E'(y)$	$E''(y)$	$-E''(x)$
$D_{3h}(1)$	A_1'	A_2'	A_2''	A_1''	$E''(x)$	$E''(y)$	$E'(x)$	$E'(y)$	A_1''	A_2''	A_2'	A_1'	$E'(x)$	$E'(y)$	$E''(x)$	$E''(y)$
C_{6v}	A_1	A_2	B_2	B_1	$E_1(y)$	$-E_1(x)$	$E_2(x)$	$E_2(y)$	A_2	A_1	B_1	B_2	$E_1(x)$	$E_1(y)$	$E_2(y)$	$-E_2(x)$
D_6	A_1	A_2	B_1	B_2	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$	A_1	A_2	B_1	B_2	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$
C_{6h}	A_g	A_g	B_g	B_g	$E_{1g}(x)$	$E_{1g}(y)$	$E_{2g}(x)$	$E_{2g}(y)$	A_u	A_u	B_u	B_u	$E_{1u}(x)$	$E_{1u}(y)$	$E_{2u}(x)$	$E_{2u}(y)$
$C_{6h}(c)$	A_g	A_g	B_g	B_g	$E_{1g}^{(a)}$	$E_{1g}^{(b)}$	$E_{2g}^{(a)}$	$E_{2g}^{(b)}$	A_u	A_u	B_u	B_u	$E_{1u}^{(a)}$	$E_{1u}^{(b)}$	$E_{2u}^{(a)}$	$E_{2u}^{(b)}$
C_{3h}	A'	A'	A''	A''	$E''(x)$	$E''(y)$	$E'(x)$	$E'(y)$	A''	A''	A'	A'	$E'(x)$	$E'(y)$	$E''(x)$	$E''(y)$
$C_{3h}(c)$	A'	A'	A''	A''	$E''^{(a)}$	$E''^{(b)}$	$E'^{(a)}$	$E'^{(b)}$	A''	A''	A'	A'	$E'^{(a)}$	$E'^{(b)}$	$E''^{(a)}$	$E''^{(b)}$
C_6	A	A	B	B	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$	A	A	B	B	$E_1(x)$	$E_1(y)$	$E_2(x)$	$E_2(y)$
$C_6(c)$	A	A	B	B	$E_1^{(a)}$	$E_1^{(b)}$	$E_2^{(a)}$	$E_2^{(b)}$	A	A	B	B	$E_1^{(a)}$	$E_1^{(b)}$	$E_2^{(a)}$	$E_2^{(b)}$
D_{3d}	A_{1g}	A_{2g}	A_{2g}	A_{1g}	$E_g(x)$	$E_g(y)$	$E_g(x)$	$E_g(y)$	A_{1u}	A_{2u}	A_{2u}	A_{1u}	$E_u(x)$	$E_u(y)$	$E_u(x)$	$E_u(y)$
$D_{3d}(1)$	A_{1g}	A_{2g}	A_{1g}	A_{2g}	$E_g(x)$	$E_g(y)$	$E_g(y)$	$-E_g(x)$	A_{1u}	A_{2u}	A_{1u}	A_{2u}	$E_u(x)$	$E_u(y)$	$E_u(y)$	$-E_u(x)$
C_{3v}	A_1	A_2	A_2	A_1	$E(y)$	$-E(x)$	$E(y)$	$-E(x)$	A_2	A_1	A_1	A_2	$E(x)$	$E(y)$	$E(x)$	$E(y)$
$C_{3v}(1)$	A_1	A_2	A_1	A_2	$E(y)$	$-E(x)$	$E(x)$	$E(y)$	A_2	A_1	A_2	A_1	$E(x)$	$E(y)$	$E(y)$	$-E(x)$
D_3	A_1	A_2	A_2	A_1	$E(x)$	$E(y)$	$E(x)$	$E(y)$	A_1	A_2	A_2	A_1	$E(x)$	$E(y)$	$E(x)$	$E(y)$
$D_3(1)$	A_1	A_2	A_1	A_2	$E(x)$	$E(y)$	$E(y)$	$-E(x)$	A_1	A_2	A_1	A_2	$E(x)$	$E(y)$	$E(y)$	$-E(x)$
C_{3i}	A_g	A_g	A_g	A_g	$E_g(x)$	$E_g(y)$	$E_g(x)$	$E_g(y)$	A_u	A_u	A_u	A_u	$E_u(x)$	$E_u(y)$	$E_u(x)$	$E_u(y)$
$C_{3i}(c)$	A_g	A_g	A_g	A_g	$E_g^{(a)}$	$E_g^{(b)}$	$E_g^{(a)}$	$E_g^{(b)}$	A_u	A_u	A_u	A_u	$E_u^{(a)}$	$E_u^{(b)}$	$E_u^{(a)}$	$E_u^{(b)}$
C_3	A	A	A	A	$E(x)$	$E(y)$	$E(x)$	$E(y)$	A	A	A	A	$E(x)$	$E(y)$	$E(x)$	$E(y)$
$C_3(c)$	A	A	A	A	$E^{(a)}$	$E^{(b)}$	$E^{(a)}$	$E^{(b)}$	A	A	A	A	$E^{(a)}$	$E^{(b)}$	$E^{(a)}$	$E^{(b)}$